

FORMULA

If a number, a , raised to a power, n , is then multiplied by a raised to another power, m , the result is $a^n a^m = a^{n+m}$.

EXAMPLE 3.107**Using the Product Rule for Exponents**

If possible, use the product rule to simplify the following:

1. $21^9 \times 21^{15}$
2. $5^9 \times 8^4$

✓ Solution

1. We can apply the product rule to simplify the expression because the bases are the same and we are multiplying.

$$21^9 \times 21^{15} = 21^{(9+15)} = 21^{24}$$
2. Since the bases are not the same (one is 5, the other 8), this cannot be simplified using the product rule for exponents.

> YOUR TURN 3.107

If possible, use the product rule to simplify the following:

1. $12^{13} \times 12^8$
2. $3^6 \times 4^{10}$

These rules can be applied to unknowns too.

EXAMPLE 3.108**Using the Product Rule for Exponents of Unknowns**

Use the product rule to simplify $a^4 \times a^{10}$.

✓ Solution

The bases are the same, and we are multiplying, so we apply the multiplication rule to simplify the expression.

$$a^4 \times a^{10} = a^{(4+10)} = a^{14}$$

> YOUR TURN 3.108

1. Use the product rule to simplify $b^6 \times b^3$.

Quotient Rule for Exponents

The next rule we examine is the quotient, or division, rule.

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When a number, a , raised to a power, n , is divided by a raised to another power, m , then the result is $\frac{a^n}{a^m} = a^{(n-m)}$.

This rule means that when we divide a base raised to a power by the same base to another power, the result is the base raised to the difference of the powers. To demonstrate, consider $\frac{14^{13}}{14^6}$. If we apply the quotient rule to that, we get $\frac{14^{13}}{14^6} = 14^{13-6} = 14^7$. This can be tested by looking at the division that is represented. Remember, 14^{13} is 14 multiplied

to itself 13 times, while 14^6 is 14 multiplied to itself 6 times. Substituting those into $\frac{14^{13}}{14^6}$ gives the following:

$$\frac{4^{13}}{4^6} = \frac{4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4}{4 \times 4 \times 4 \times 4 \times 4 \times 4}$$

We see here that there are a LOT of fours to be divided out.

$$= \frac{\cancel{4} \times \cancel{4} \times \cancel{4} \times \cancel{4} \times \cancel{4} \times \cancel{4} \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4}{\cancel{4} \times \cancel{4} \times \cancel{4} \times \cancel{4} \times \cancel{4} \times \cancel{4}} = \frac{4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4}{1} = 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4$$

What remains is 4 to the 7th power, $4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 = 4^7$.

All of the work above confirmed what the formula told us would be the result.

 *Caution: The quotient rule only applies when the bases are the same. If the bases are different, we do not apply this rule.*

EXAMPLE 3.109

Using the Quotient Rule for Exponents

Use the quotient rule to simplify $\frac{5^{19}}{5^{11}}$.

Solution

We can apply the quotient rule to simplify the expression since the bases are the same and we are dividing.

$$\frac{5^{19}}{5^{11}} = 5^{(19-11)} = 5^8$$

YOUR TURN 3.109

1. Use the quotient rule to simplify $\frac{b^6}{b^4}$.

VIDEO

[Product and Quotient Rule for Exponents \(https://openstax.org/r/Product_and_Quotient_Rule_for_Exponents\)](https://openstax.org/r/Product_and_Quotient_Rule_for_Exponents)

A natural consequence of the quotient rule is what it means to raise a non-zero number to the zeroth power. Let's look at the simplification when the exponents are equal.

$$\frac{3^6}{3^6} = 3^{(6-6)} = 3^0$$

We know that a number divided by itself is 1, so $\frac{3^6}{3^6} = 1$. From that it must be that $\frac{3^6}{3^6} = 3^0 = 1$. This provides the rule for a number raised to the power 0: $a \neq 0$.

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If you have a non-zero number a , then $a^0 = 1$.

Distributive Rule for Exponents

The next rule we look to is a distributive rule for exponents.

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If you have a product, $(a \times b)$, and raise it to an exponent, n , then $(a \times b)^n = a^n \times b^n$.

This means that when we have two numbers multiplied together, and that is raised to a power, it is the same as raising each of the numbers to the same power first, then multiplying. For example, $(3 \times 7)^4 = 3^4 \times 7^4$. This can be explained using the definition of exponents and multiplying all the factors.

$$(3 \times 7)^4 = (3 \times 7) \times (3 \times 7) \times (3 \times 7) \times (3 \times 7)$$

We may change the order in which numbers are multiplied. This is the commutative property of the real numbers. This can be written as $3 \times 3 \times 3 \times 3 \times 7 \times 7 \times 7 \times 7$. Using exponents, that shortens to $3^4 \times 7^4$.

This also works in the other direction, $a^n \times b^n = (a \times b)^n$. Read this way, if we have one base raised to an exponent, and another base raised to the same exponent, we can multiply the bases and raise that product to the shared exponent. For instance, $7^8 \times 11^8 = (7 \times 11)^8 = 77^8$.

 *Caution: The exponent distributive rule, $a^n \times b^n = (a \times b)^n$, only works if the exponents are the same.*

EXAMPLE 3.110**Using the Distributive Rule for Exponents**

Use the exponent distributive rule to expand $(6 \times 13)^7$.

 Solution

Applying the distributive rule to the product, we get $(6 \times 13)^7 = 6^7 \times 13^7$.

 YOUR TURN 3.110

1. Use the exponent distributive rule to expand $(2 \times 19)^{14}$.

EXAMPLE 3.111**Using the Distributive Rule for Exponents**

Use the exponent distributive rule to expand $(c \times d)^{10}$.

 Solution

Applying the distributive rule to the product, we get $(c \times d)^{10} = c^{10} \times d^{10}$.

 YOUR TURN 3.111

1. Use the exponent distributive rule to expand $(a \times b)^6$.

This distribution also works for quotients. A fraction raised to an exponent equals the numerator raised to the exponent divided by the denominator raised to the exponent. For example, $\left(\frac{3}{5}\right)^7 = \frac{3^7}{5^7}$. Demonstrating this is similar to the previous rule.

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When you have a fraction, $\frac{a}{b}$, raised to an exponent, n , then $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$.

EXAMPLE 3.112**Using the Distributive Rule for Exponents with Fractions**

Use the exponent distributive rule to expand the following:

1. $\left(\frac{4}{9}\right)^6$
2. $\left(\frac{3}{b}\right)^{11}$

✓ Solution

1. Applying the distributive rule to the quotient, we get $\left(\frac{4}{9}\right)^6 = \frac{4^6}{9^6}$.
2. Applying the distributive rule to the quotient, we get $\left(\frac{3}{b}\right)^{11} = \frac{3^{11}}{b^{11}}$.

> YOUR TURN 3.112

Use the exponent distributive rule to expand the following:

1. $\left(\frac{14}{5}\right)^9$
2. $\left(\frac{a}{18}\right)^5$

▶ VIDEO

Fraction Raised to a Power (https://openstax.org/r/Fraction_Raised_to_a_Power)

Power Rule

In the previous two sets of rules, we've seen exponents applied to products and quotients. Now we look to exponents applied to other exponents. For example, $(3^6)^4 = 3^{(6 \times 4)} = 3^{24}$. This can be explained by examining what the outer exponent does. We raise 3^6 to the fourth power, so we multiply 3^6 by itself 4 times, $(3^6)^4 = 3^6 \times 3^6 \times 3^6 \times 3^6$. Now if we apply the product rule for exponents, this becomes $3^{(6+6+6+6)} = 3^{24}$.

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If you raise a non-zero base, say a , to an exponent n , and raise that to another exponent, m , you get the base raised to the product of the exponents, which is $(a^n)^m = a^{(n \times m)}$.

EXAMPLE 3.113**Raising an Exponent to an Exponent**

Expand the following:

1. $(6^7)^3$
2. $(b^{12})^4$

✓ Solution

1. Using the power rule of exponents, $(6^7)^3 = 6^{(7 \times 3)} = 6^{21}$.
2. Using the power rule of exponents, $(b^{12})^4 = b^{(12 \times 4)} = b^{48}$.

> YOUR TURN 3.113

Expand the following:

1. $(11^4)^{12}$
2. $(a^7)^6$

Negative Exponent Rule

Up until now, we've only looked at positive exponents. The last exponent rule we look at is what negative exponents represent. Recall the quotient rule: $\frac{a^n}{a^m} = a^{(n+m)}$. What would happen if the exponent in the denominator was larger

than that in the numerator? For example, $\frac{4^5}{4^7}$. If we apply the quotient rule, we obtain $\frac{4^5}{4^7} = 4^{5-7} = 4^{-2}$. We need to make sense of that negative exponent. To do so, we can expand the quotient and see what happens:

$\frac{4^5}{4^7} = \frac{4 \times 4 \times 4 \times 4 \times 4}{4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4}$. When we divide out common factors, only two factors of 4 are left in the denominator, as we see here: $\frac{1}{4 \times 4}$. Using exponent notation, this is $\frac{1}{4^2}$. Since 4^{-2} and $\frac{1}{4^2}$ represent the same number, $\frac{4^5}{4^7}$, they are equal. This demonstrates how negative exponents are defined.

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$$a^{-n} = \frac{1}{a^n} \text{ provided that } a \neq 0.$$

$$\text{Similarly, } \frac{1}{a^{-n}} = a^n.$$

EXAMPLE 3.114

Eliminating Negative Exponents

Convert the following to expressions with no negative exponent:

1. $3^4 \times 5^{-8}$
2. $a^{-9} \times b^5$
3. $\frac{7}{c^{-2}}$

Solution

1. Using the negative exponent rule on the 5^{-8} and multiplying, $3^4 \times 5^{-8} = 3^4 \times \frac{1}{5^8} = \frac{3^4}{5^8}$.
2. Using the negative exponent rule on the a^{-9} and multiplying, $a^{-9} \times b^5 = \frac{1}{a^9} \times b^5 = \frac{b^5}{a^9}$.
3. Begin by rewriting the expression as $\frac{7}{c^{-2}} = \frac{7}{1} \times \frac{1}{c^{-2}}$. Apply the negative exponent rule to $\frac{1}{c^{-2}}$ in the expression, which becomes $\frac{7}{1} \times \frac{1}{c^{-2}} = 7 \times c^2$, which has no negative exponents.

YOUR TURN 3.114

Convert the following to expressions with no negative exponent:

1. $12^{-3} \times 7^5$
2. $c^{-7} \times 5^3$

EXAMPLE 3.115

Eliminating Denominators by Using Negative Exponents

Use negative exponents to rewrite the following expressions with no denominator:

1. $\frac{7^3}{13^9}$
2. $\frac{c^4}{d^8}$

 Solution

1. Rewrite the expression $\frac{7^3}{13^9}$ as $\frac{7^3}{1} \times \frac{1}{13^9}$. Then use the definition of negative exponents to rewrite the $\frac{1}{13^9}$ as 13^{-9} .
Last, multiply, yielding $\frac{7^3}{1} \times \frac{1}{13^9} = 7^3 \times 13^{-9}$.
2. Rewrite the expression $\frac{c^4}{d^8}$ as $\frac{c^4}{1} \times \frac{1}{d^8}$. Then use the definition of negative exponents to rewrite the $\frac{1}{d^8}$ as d^{-8} .
Last, multiply, yielding $\frac{c^4}{1} \times \frac{1}{d^8} = c^4 \times d^{-8}$.

 YOUR TURN 3.115

Use negative exponents to rewrite the following expressions with no denominator:

1. $\frac{6^3}{13^8}$
2. $\frac{c^5}{2^9}$

The table below shows a summary of the exponent rules from this section.

Rule	Example	In Words
Product Rule $a^n a^m = a^{n+m}$	$8^2 \times 8^5 = 8^7$	A base raised to a power, times the same based raised to another power, is the base raised to the sum of the powers.
Quotient Rule $\frac{a^n}{a^m} = a^{(n-m)}$	$\frac{11^{14}}{11^{12}} = 11^2$	A base raised to a power, divided by the same based raised to another power, is the base raised to the difference of the powers.
Zero Power Rule $a^0 = 1$ provided that $a \neq 1$	$412^0 = 1$	Any non-zero number raised to the zeroth power equals 1.
Distributive Rule, Multiplication $(a \times b)^n = a^n \times b^n$	$(14 \times 31)^9 = 14^9 \times 31^9$	Exponents distribute across multiplication.
Distributive Rule, Division $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	$\left(\frac{62}{91}\right)^8 = \frac{62^8}{91^8}$	Exponents distribute across division.
Power Rule $(a^n)^m = a^{(n \times m)}$	$(5^9)^{15} = 5^{135}$	A base raised to a power, raised to another power, is the base raised to the first power times the second power.
Negative Exponent Rule $a^{-n} = \frac{1}{a^n}$ provided that $a \neq 0$	$6^{-8} = \frac{1}{6^8}$ $\frac{1}{12^7} = 12^{-7}$	A base raised to a negative exponent is 1 divided by the base raised to the positive power, and vice versa.

These rules often occur in tandem with each other, but it requires that you carefully apply the rules.

EXAMPLE 3.116

Simplifying Expressions Using Exponent Rules

Simplify the following:

1. $\left(\frac{4^2 \times 7}{9^3}\right)^5$
2. $\left(\frac{5a^4}{b^9}\right)^6$

 Solution

1. **Step 1:** To simplify this, we start by distributing the power 5 across the quotient:

$$\left(\frac{4^2 \times 7}{9^3}\right)^5 = \frac{(4^2 \times 7)^5}{(9^3)^5}$$

Step 2: We distribute the power 5 in the numerator across that multiplication:

$$\left(\frac{4^2 \times 7}{9^3}\right)^5 = \frac{(4^2 \times 7)^5}{(9^3)^5} = \frac{(4^2)^5 \times 7^5}{(9^3)^5}$$

Step 3: We apply the power rule where indicated:

$$\left(\frac{4^2 \times 7}{9^3}\right)^5 = \frac{(4^2 \times 7)^5}{(9^3)^5} = \frac{(4^2)^5 \times 7^5}{(9^3)^5} = \frac{4^{(2 \times 5)} \times 7^5}{9^{(3 \times 5)}} = \frac{4^{10} \times 7^5}{9^{15}}$$

2. **Step 1:** To simplify this, we start by distributing the power 6 across the quotient:

$$\left(\frac{5a^4}{b^9}\right)^6 = \frac{(5a^4)^6}{(b^9)^6}$$

Step 2: We distribute the power 6 in the numerator across that multiplication:

$$\frac{(5a^4)^6}{(b^9)^6} = \frac{(5)^6 \times (a^4)^6}{(b^9)^6}$$

Step 3: We apply the power rule where indicated:

$$\frac{(5)^6 \times (a^4)^6}{(b^9)^6} = \frac{5^6 a^{24}}{b^{54}}$$

 YOUR TURN 3.116

Simplify the following:

1. $\left(\frac{7^9}{10^5 \times 6^3}\right)^8$
2. $\left(\frac{4}{a^9 b^6}\right)^2$

 VIDEO

[Simplifying Expressions with Exponents \(https://openstax.org/r/Simplifying_Expressions_with_Exponents\)](https://openstax.org/r/Simplifying_Expressions_with_Exponents)

Check Your Understanding

40. Simplify $a^3 \times a^5$.
41. Simplify $\frac{5^4}{5^8}$.
42. Simplify $(6b)^9$.
43. Simplify $\left(\frac{c}{7}\right)^3$.

44. Simplify $\left(\frac{3a^2}{4b^5}\right)^6$.



SECTION 3.8 EXERCISES

For the following exercises, simplify the expression.

1. $4^5 \times 4^2$
2. $3^3 \times 3^6$
3. $a^2 \times a^7$
4. $b^7 \times b^{12}$
5. $\frac{4^6}{4^2}$
6. $\frac{15^{14}}{15^{11}}$
7. $\frac{c^9}{c^4}$
8. $\frac{a^{14}}{a^5}$
9. $\frac{11^3}{11^7}$
10. $\frac{7^{14}}{7^{23}}$
11. $\frac{b^6}{b^{19}}$
12. $\frac{d^8}{d^{17}}$
13. $(4 \times 3)^4$
14. $(5 \times 8)^7$
15. $(3c)^6$
16. $(n \times m)^9$
17. $\left(\frac{7}{2}\right)^8$
18. $\left(\frac{a}{6}\right)^{11}$
19. $\left(\frac{4}{c}\right)^9$
20. $(6^4)^8$
21. $(12^5)^3$
22. $(x^6)^2$
23. $(b^6)^{11}$
24. $\left(\frac{3b}{5}\right)^4$
25. $\left(\frac{a}{2b}\right)^5$
26. $\left(\frac{3a^4}{7}\right)^3$
27. $\left(\frac{6x^7}{11}\right)^8$
28. $\left(\frac{2x^4}{b^7}\right)^3$
29. $\left(\frac{7r^9}{a^4}\right)^{12}$

For the following exercises, rewrite the expression without a denominator.

30. $\frac{3}{a^4}$
31. $\frac{5}{b^6}$
32. $\frac{4}{b^2c^3}$
33. $\frac{9}{x^4y^5}$

For the following exercises, rewrite the expression without negative exponents.

34. $12^4 \times 5^{-3}$
35. $3b^{-12}$